Motion Generation in MR Environment Using Electrostatic Film Motor for Motion-Triggered Cine-MRI

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***Abstract*—This paperproposes a new methodology of biomechanical modeling for visualizing tissue deformation, combining an electrostatic film motor and tagged cine-MR imaging . The electrostatic film motor has a simple structure, is easily controlled by open loop, and its MR-compatibility has recently been verified. Tagged cine-MRI is an established process for measuring deformation of active objects. The newly proposed methodology enables the application of cine-MR imaging to passive objects. The electrostatic motor is used to create deformation on passive soft samples inside an MR scanner, the internal deformation of which is visualized by tagged cine-MR imaging. The force applied on the samples is measured by a force sensor fabricated on shielded strain gauges. The impact of the developed system on the MR imaging is verified through SNR evaluation. Then, the proposed methodology is applied to visualizing deformations of a gel sample and a human upper arm to verify the applicability of this methodology in biomechanical measurements.**

***Index Terms*—Biomechanics, electrostatic motor, MR-compatibility, shielded strain gauge, tagged cine-MRI.**

I. Introduction

In RECENT years, mathematical modeling of biomechanical structures has become a popular research topic, as the mathematical models can be used for verifying the mechanism of pathological lesion and the skeletal muscular changes due to senile decay [1]. Although researchers have succeeded in partial elucidation of the muscular mechanism, no *de facto* standard for biomechanical modeling has been established yet. Analyzing topographic images, especially MR images, of organs is one of the major approaches in the field of biomechanics. In this paper, we describe our effort to establish a new methodology of biomechanical MR imaging that combines an electrostatic motor and tagged cine-MR imaging.

The conventional method in MRI-related biomechanical modeling is obtaining static images of both an initial state and a stressed state of a target sample and calculating the difference in volume distribution to verify the biomechanical properties such as muscular elasticity [2], [3]. The advantage of this method is its simplicity; however, it does not allow verifying the dynamic deformation of organs. To visualize the dynamic deformation, it is necessary to obtain continuous frames of image during the deformation. For such continuous scanning, two imaging techniques are mainly used: echo planner imaging (EPI) and trigger synchronized imaging. EPI is a rapid imaging method that allows obtaining a series of images in short time [4]. It has high time resolution; however, the spatial resolution is not high enough to use the images for mathematical modeling. The more popular method in biomechanical modeling would be trigger synchronized imaging, which is often mentioned as cine-MRI.

There have been reported various studies related to biomechanical modeling using cine-MR imaging, e.g., visualization of cardiac motion [5], a knee joint motion [6], and tongue motion during speech [7], [8]. For visualizing dynamic motions, there should be a driving power source to generate motions. In the aforementioned studies, the driving sources were muscles of the subjects. In that case, the application of the technique is limited to organs that can be actuated by muscles. To realize deformation for dynamic visualization in static organs that cannot be moved by muscles, an external driving mechanism, such as electric motors, becomes necessary.

There are, however, significant challenges for conventional electric motors to work within an MRI environment [9], [10]. The extraordinary magnetic field of MR scanners prevents the use of electromagnetic actuators, which could easily be attracted by the field and could possibly cause fatal accidents (which is often referred to as "missile effect"). Moreover, magnets and the ferromagnetic materials used in the actuators interfere with the scanner's magnetic field and scanning RF pulses, causing artifacts in the resultant MR images.

To overcome these difficulties, various alternative actuators have been investigated [10], including ultrasonic motors [11], [12], hydraulic/pneumatic actuators (or transmissions) [13], [14], and electrostrictive polymer actuators (EPAM) [15]. Among these, ultrasonic actuators have been primarily studied in the previous studies [10]. For example, they have been employed in manipulators of MRI-guided surgical robots [11], which have exhibited good MR-compatibility in some aspects. Ultrasonic actuators, however, are not fully MR-compatible at this moment because they were reported to disturb MRI if they are operated during MR scanning [12]. In the aforementioned

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cine-MRI technology, motion control must be performed during scanning, thus requiring another MR-compatible actuator that can operate together with MR scanning.

In some recent studies, MR-compatible actuators that can work together with scanning have been reported, which includes hydraulic power transmission and piezoelectric actuators [16], [17]. These actuators are designed for phase contrast (PC) imaging and/or MR elastography. Because of the different requirement, their motion stroke is far smaller, a few millimeters or less, which is too small to realize noticeable deformation on organs. To realize noticeable deformation in, e.g., human upper arm, motion stroke more than 1 cm would be required.

In this paper, we propose a new methodology of cine-MR imaging, combining an electrostatic film motor and tagged cine-MR process. The core component in our proposal is a high-power electrostatic linear motor [18], the MR-compatibility of which has already been verified [19]. Since the electrostatic motor is a synchronous actuator, motion control can be realized in an open-loop manner without a displacement sensor. Since the use of a displacement sensor can cause another MR-compatibility problem, the capability of open-loop motion control is an advantage in MR environments.

In the next section, the paper describes the cine-MR imaging technique with which the actuator is to be combined. In Section III, the overview of the actuator is described, followed by the detailed description of supplemental components such as a force sensor. Section IV describes the results of experiments applying the system for visualizing deformation of a gel sample and a human arm.

II. Target Application

The target application of the proposed methodology is biomechanical modeling, in which the distribution of viscoelastic parameters in a target object will be identified. For this purpose, the MRI will take sequential images of dynamic deformation of a target object. This section describes key technological components in the methodology.

A. Trigger Synchronized Imaging

Cine-MRI is a type of imaging technique that allows sequential imaging of moving objects. Fig. 1 illustrates the principle of motion-triggered cine imaging. In cine-MRI, a repetition of the same motion, which is synchronized to MR scanning, is necessary to obtain sequential images [20]. Cine-MRI acquires image data over many cycles of a repeating motion and then retrospectively sorts the data into sequential images [5], [20]. The technique was originally developed for visualization of cardiac motion [5], but now widely used for visualizing moving organs [6]–[8], [16].

Since cine-MRI technique collects data over many motion cycles, the motion should be repeated many times (i.e., over 100 times) with a constant cycle and considerably similar motion profiles. In addition, the MR scanning must be synchronized with the repeating motion. Therefore, in cine-MR imagingof cardiac motions, electrocardiogram (ECG) pulse is measured from the subject, and used as the trigger pulse for scanning. For

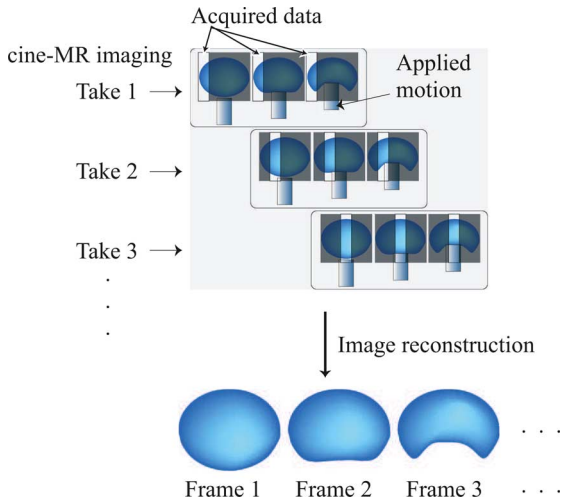


Fig. 1. Principle of motion-triggered cine-MRI (image diagram).

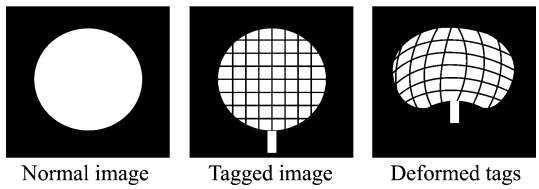


Fig. 2. Image diagram of tags and tag deformation.

the motions other than cardiac, such as knee motion and tongue motion, artificially generated trigger pulse was fed both to MR scanner and to the subject within the scanner. The subject repeats the required motions, such as knee bending [6] or pronouncing a sound [7], [8], based on the trigger signal.

B. Tagging

For modeling of organs, "tagging" is sometimes used with cine-MRI technique. Tagging is a technique to generate lines or grid patterns on the sample image by producing a periodic spatial modulation of magnetization (SPAMM) prior to imaging [21], [22]. Fig. 2 illustrates the output of tagged images, and tag deformation. Motion occurring during MR scanning will be reflected as a displacement of the tags [21]. Tagged cine-MRI, a combination of this technique and cine-MRI, allows noninvasive visualizing of the internal deformation of a moving tissue by tracking the tagged region.

C. Motion Generation

MR-compatible actuating system is necessary to generate motion on a static sample in MR environment. To combine the external driving system with cine-MRI process, the driving system should be capable of generating constant cyclic motion that is synchronized with MR scanning. Although such motion control can be easily realized by feedback control, it typically requires additional displacement (or velocity) sensors that can raise another MR-compatibility problem (most sensors are not MR-compatible). Hence, in this scheme, selection of an actuator is one of the most important issues. Here, we propose to utilize an electrostatic film motor. Since the motor is a synchronous drive,

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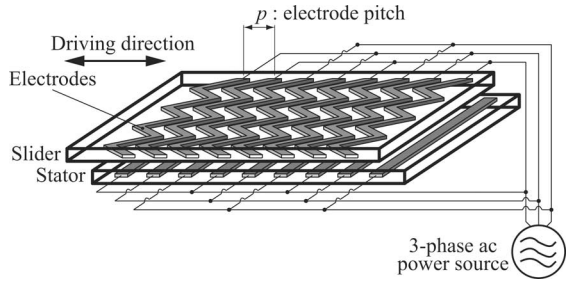


Fig. 3. Basic structure of the electrostatic motor.

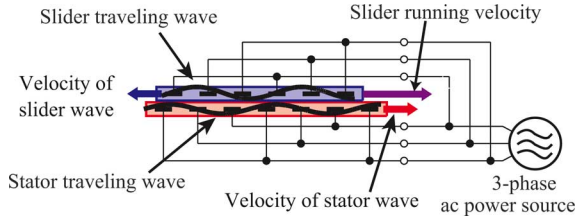


Fig. 4. Driving principle of the electrostatic motor.

it can easily generate the constant repeating motion without the use of external position sensors, the details of which are described in the next section.

III. Electrostatic Motor System for Motion Generation

This section first describes the basic principle of the electrostatic film motor, which is followed by the explanation of the developed force sensor that is used in combination with the motor. In order to verify the applicability of the combination of the motor and the sensor for the aforementioned cine-MR process, we conducted some preliminary tests in normal environment, the details of which are described in the latter part of this section.

A. High-Power Electrostatic Linear Motor

1) Principle and Basic Structure: The electrostatic film motor used in this study is *dual excitation multiphase electrostatic drive* [18], [19]. Differently from other electrostatic motors, this motor can generate practical large thrust force, such as several newtons. The basic structure of the motor is shown in Fig. 3. In its simplest configuration, the motor consists of a pair of thin plastic films that are manufactured using flexible printed circuit (FPC) technology. Each film contains three-phase electrodes that are aligned at regular 200-μm intervals. In our experiment, we used "skewed electrodes" to reduce the force ripple [19], [23]. Fig. 4 illustrates the driving principle of the motor. The motor is driven with a three-phase ac high voltage. As depicted in the figure, the three-phase voltage source is connected to the three-phase electrodes on both the stator and the slider, but with the connection order reversed. The motor motion is synchronized to the frequency (or phase) of the applied ac; the resultant motor speed is proportional to the applied frequency (or the motor displacement is proportional to the phase of applied ac). However, if excessive external force is applied to the slider, the motor will "step-out," which means to get out of synchronization. This force limit depends on the amplitude of the applied ac voltage and is proportional to the square of the amplitude.

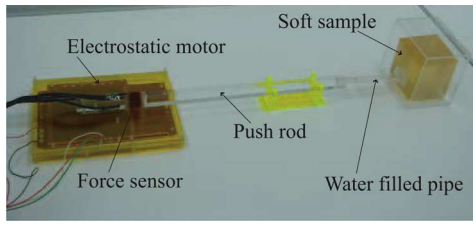


Fig. 5. Experimental setup in normal environment.

2) Prototype for the Experiments: Fig. 5 shows the experimental setup. The motor consists of two sets of stator and slider films for larger output force capability. The films measure 18.1 cm $\times$ 10.8 cm and 11.4 cm $\times$ 12.8 cm, respectively, and their thickness is approximately 200 μm. The two sets of films are stacked in a plastic box that is filled with Fluorinert to prevent electric breakdown of air. The slider films are screwed to a plastic slider board on which a 50-cm-long rod is connected. The other end of the rod is filled with water so that it will appear on MR images. On the tip of the rod, round plate is attached so that the pushing force to the sample will be distributed over a large area. In a preliminary test, the limit of the external force for preserving synchronous driving was approximately 10 N when ac voltage of 1.6 kV_{0-p} was applied.

B. Force Sensor

For biomechanical modeling, the force applied to the sample must be recorded during the deformation. Since the sensor is used in an MRI environment during scanning, it should also be MR-compatible. Such MR-compatible force sensors were reported in some literature [24], [25]; however, there is no commercially available one. To combine with the electrostatic motor, we fabricated a force sensor using commercially available strain gauges.

The strain gauges used in our study is shielded foil strain gauge (KFS-5-350-J1, Kyowa Electronic Instruments). A pair of gauges was bonded orthogonally to mutually eliminate the induced noise on gauge electrodes. Two pairs of gauges were bonded to the push-rod, as shown in Fig. 6. The usable range of this sensor in MR environment was evaluated in our previous work, which is more than 35 cm from the center of the magnetic field [26].

C. Applicability Tests for Sample Deformation

To combine the electrostatic motor with cine-MR process, it should perform cyclic motion without step-out to ensure constant repeating motions. If the resistive force from a target object is too large, the motor can step-out from the synchronization and the resultant motor motion becomes not constant. Therefore, we conducted an experiment in normal environment using the same

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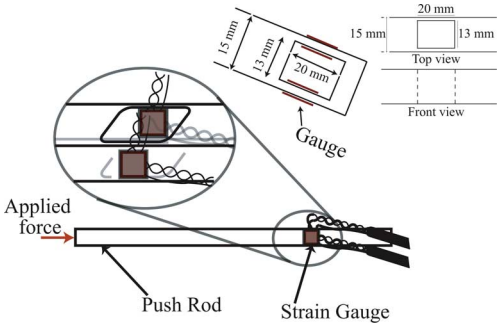


Fig. 6. Fabricated force sensor. Four KFS strain gauges bonded on the push-rod. Two orthogonally aligned gauges are bonded on both sides of the rod.

soft sample and the same motion profile, and measured the motor motion to ensure that there is no step-out.

Three high-voltage amplifiers (AS-123, NF crop) are used as the power source for the motor. A waveform signal generator (AG-1200, Yokogawa electric) generated a three-phase ac signal that was fed to the high-voltage amplifiers to be thousand-folded before being applied to the motor. Geletin gel is used as the target sample in this experiment. The gel was fixed on a table, as shown in Fig. 5. While push-rod is pushing the gel, a laser displacement sensor (NAiS LM10, ANR1215) measured the displacement of the tip of the push-rod. The force sensor was connected to a strain amplifier (Kyowa DPM 712B) with a low-pass filter (LPF) of 100 Hz to measure the applied force. At the same time, applied voltage and current to the motor were measured.

The motor repeated the following reciprocating motion for more than 100 times. First, the waveform generator issues a trigger signal, with a pulsewidth of 50 ms. Then, 200 ms after the trigger, the waveform generator generates 20 Hz and 1.6 kV_{0-p} ac signal for 500 ms, which corresponds to the motor motion of 24 mm/s toward the gel sample. After that, the waveform generator generates 10 Hz and 1.6 kV_{0-p} ac signal for 1000 ms, which results in backward motion of the motor at a speed of 12 mm/s. After retreating back to the original position, the motor rests for 400 ms. Resultant one cycle of the motion measures 2.15 s.

Fig. 7 shows the results. From the top, the plots show the trigger signal, displacement of the push-rod, output of the force sensor, applied voltage, and current. The displacement plot shows that the motor repeated the intended motion without step-out. This verifies that the motor is capable of generating constant cyclic motion by pushing a soft sample.

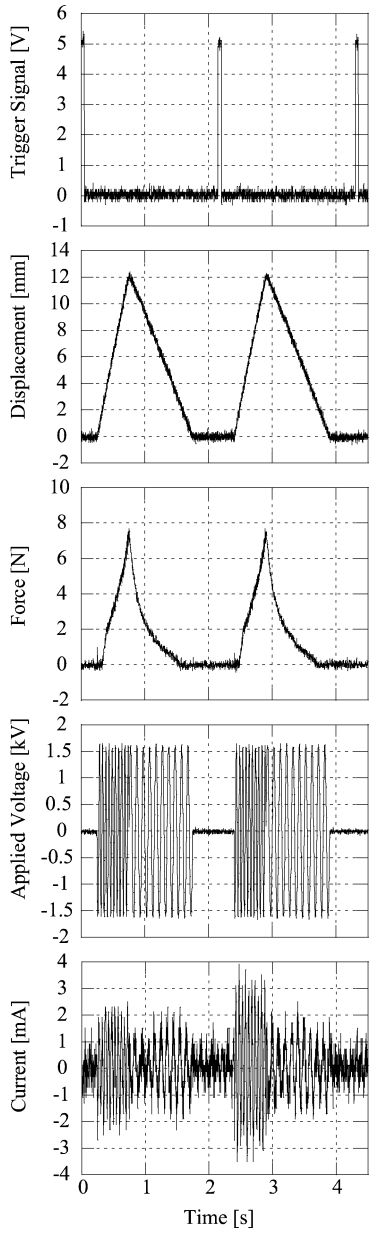


Fig. 7. Results of the experiments conducted in normal environment.

IV. Experiments in MR Environment

A. Evaluation of the SNR of MR Images

Before applying the system for deformation visualization, we first confirmed the impact of the system on MRI through SNR evaluation. The MR scanner used in the experiments is Toshiba Excelart that utilizes a 1.5 T static magnetic field for its imaging. Maximum gradient amplitude is 23 mT/m and the maximum RF pulse intensity is 20 kW.

Fig. 8 shows the schematic diagram of the experimental setup used in the MR environment. The electrostatic motor was placed on the bed of the MR scanner approximately 40 cm away from the center. On the center of the imaging region, a cubical phantom for SNR evaluation was placed, instead of a soft sample. Its coronal images were taken for SNR evaluation using a head coil.

The amplifiers and signal generator are placed outside the MR room, which are connected to the motor using 8-m-long shielded cable through the door of the MR room, which was kept slightly open to send the cables in. A choke coil was inserted between the amplifiers and the shielded cable to prevent introducing high-frequency noise into the MR room. The synchronizing trigger signal generated by the waveform generator was fed to the MR scanner through the external trigger port of the scanner.

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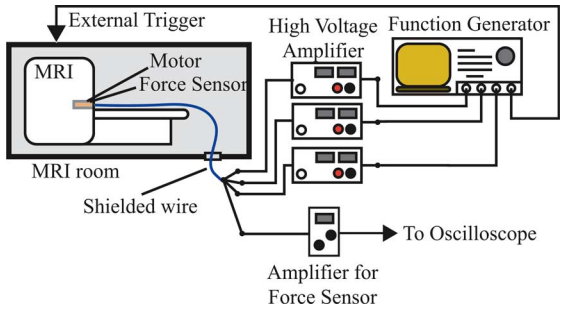


Fig. 8. Schematic diagram of the setup used in the MR environment. Amplifiers and a signal generator are placed outside the MR room. An open-loop control was used to control the motor. A trigger pulse was sent to synchronize the motor motion and the imaging process.

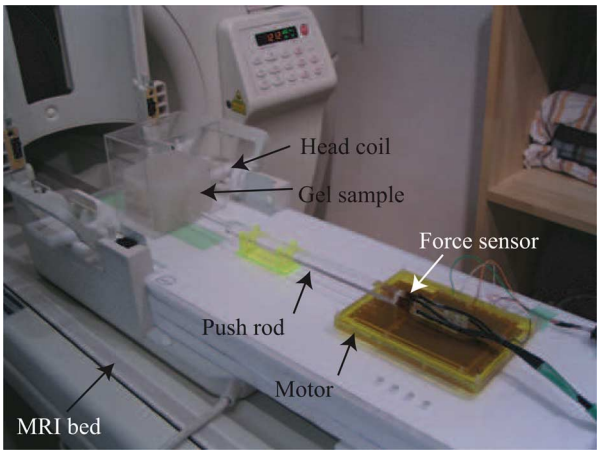


Fig. 9. Experimental setup in MR environment.

The imaging sequence used in the test was a gradient echo pulse of TR/TE = 29/15 ms. The SNR for each image was calculated as the ratio of the mean signal value at the center of the image (signal) against the standard deviation value of the four corners (noise) [9]. Images were acquired while motor is at rest and operated by an applied voltage of 1.6 kV_{0-p}. The motor was driven at 10 Hz (12 mm/s) and 20 Hz (24 mm/s) of driving frequencies. Fig. 10 shows the obtained images. Subtracted images were derived to check whether any image distortion occurred due to the presence of the motor.

The intrinsic SNR of the phantom image, which was measured without motor, was 65.8, and the SNR with motor at rest was 47.6. SNR corresponding to a driving frequency of 10 Hz was 44.9 and 45.86 for 20 Hz. This degradation of SNR would be due to the electric noise coming from the amplifiers and the motor itself, which contains conductive material (mainly copper). However, no image distortion is observed in the subtracted images.

B. Experiments Using Gel Samples

Using the developed system, we observed dynamic deformation of a gel sample. The used sample was gelatin gel with a volume of 80 cm × 80 cm × 100 cm.

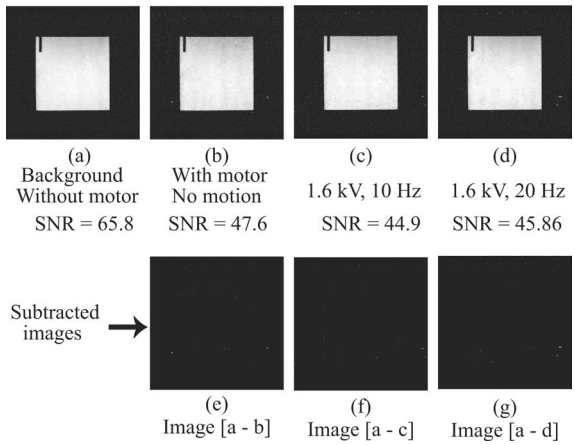


Fig. 10. Comparison of MR images of phantom corresponding to different conditions.

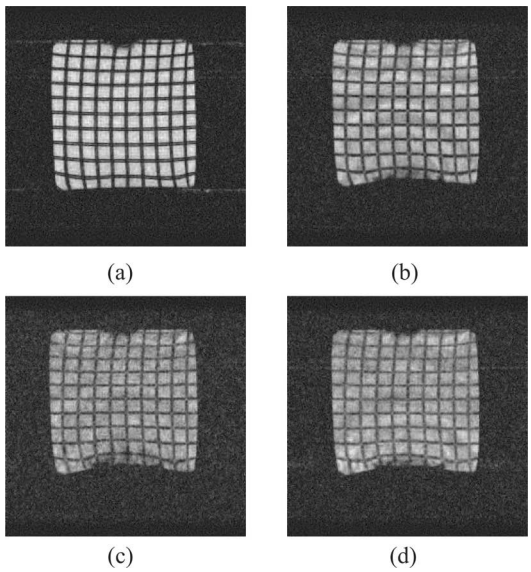


Fig. 11. Tagged cine-MR images of gelatin gel. (a) Frame 2. (b) Frame 13. (c) Frame 17. (d) Frame 20.

1) Procedure: Tagged cine-MRI process was used to visualize the deformation of the samples. Coronal surface was selected and mesh-type tags with interval of 8 mm were applied on that surface. The motor motion sequence is the same as the normal environment test described in Section III-C. For this experiment, the motion was repeated more than 256 times.

In the cine imaging settings, there was a 200 ms of delay time in each cycle after the trigger pulse until starting of the image acquisition. The imaging sequence used for obtaining sequential images is field echo sequence of TR/TE = 29/15 ms, FA = 20°, slice thickness = 8 mm, FOV = 12.8 cm, MTX = 256 × 256, delay time = 200 ms, and tag pitch = 8 mm. Twenty-five chronological images were obtained in one image acquisition.

2) Results and Discussion: Fig. 11 shows the images of the gelatin gel. Fig. 11(a) and (b) shows the intermediate deformation while pushing and Fig. 11(c) shows the maximum deformation. Fig. 11(d) shows the realignment of the gelatin gel.

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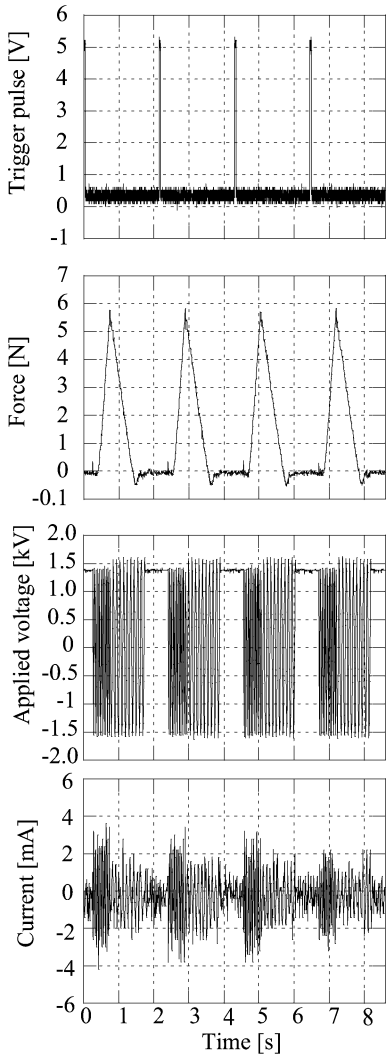


Fig. 12. Output of the force sensor, corresponding applied voltage, and current of the motor for gelatin gel.

Fig. 12 shows the output of the force sensor and the corresponding applied voltage of the motor. When the rod pushes the sample, the force increases gradually and then gradually decreases as the rod retreats to the original position. Although there should have been no pulling force on the pushing rod, the sensor output shows negative spike (meaning pulling force) at the end of each cycle. The negative output would have been caused by the bending of the rod, since it was confirmed that the fabricated sensor had some sensitivity also to the bending.

C. Visualizing Deformation of Human Upper Arm

Since we succeeded in obtaining tagged cine-MR images of gel samples, we expanded this system to obtain image data of human muscle. As a first step, the system was applied to visualize internal deformation of a human upper arm.

1) Procedure: Fig. 13 shows the experimental setup. A pair of surface coils was used to obtain images, since the head coil is hard to use for measuring the arm. The SNR of the images taken by the surface coil is smaller than that by the head coil.

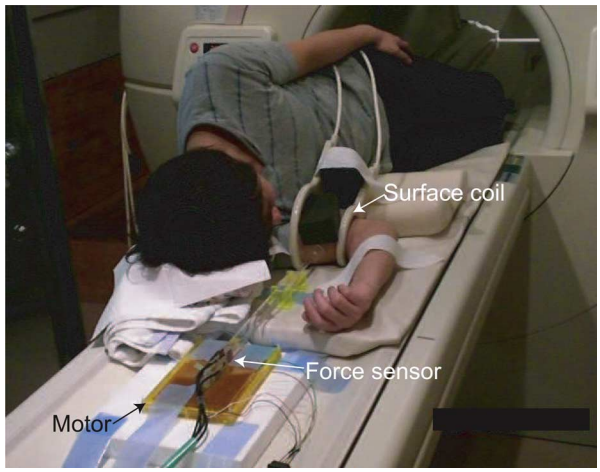


Fig. 13. Experimental setup used to obtain MR images of human upper arm.

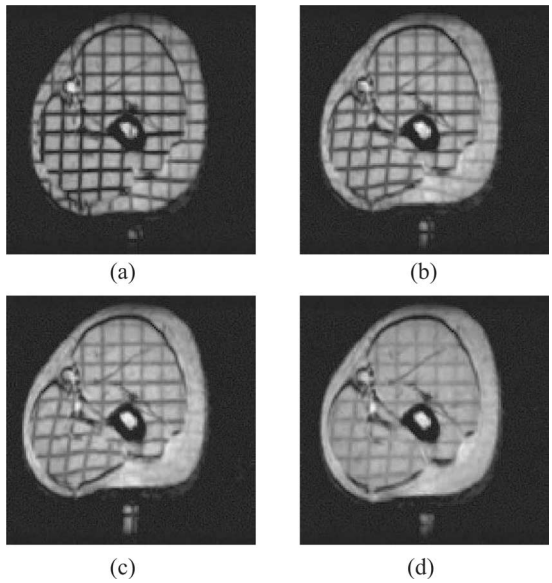


Fig. 14. Tagged cine-MR images of human upper arm. (a) Frame 1. (b) Frame 4. (c) Frame 6. (d) Frame 10.

The driving sequence of the motor is the same as that for the gel sample. Imaging sequence is field echo sequence of TR/TE = 100/15 ms, FA = 20°, slice thickness = 8 mm, FOV = 12.8 cm, MTX = 128 × 128, NAQ = 2, delay time = 200 ms, and tag pitch = 10 mm. Seventeen chronological images were obtained.

2) Results and Discussion: Fig. 14 shows the tagged cine-MR images of the upper arm: frame 1 shows the starting position, frame 4 shows an intermediate deformation point, frame 6 shows the maximum deformation, and frame 10 shows the realignment. Since the motor was moving at 24 mm/s during pushing and the TR of the imaging was 100 ms, the deformation proceeds at the speed of 2.4 mm/s. Therefore, the resultant images may have uncertainty of 2.4 pixels in the maximum. However, the images clearly showed the deformation of organs and tags.

Fig. 15 shows the output of the force sensor for four motion cycles, corresponding applied voltage, and current of the motor. Although the force measurement contains some spike noises,

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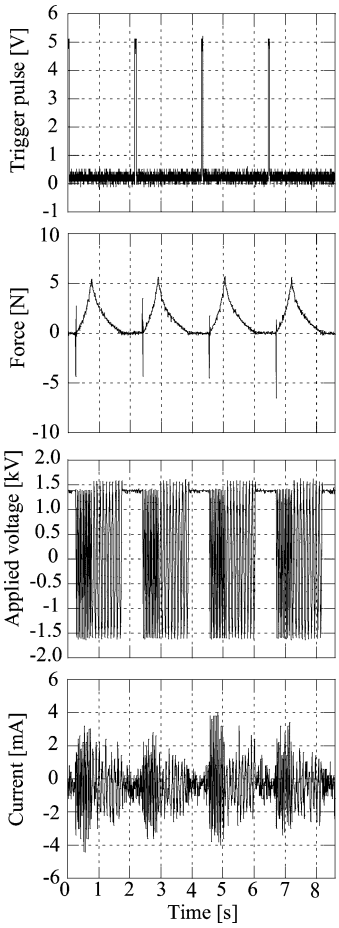


Fig. 15. Output of the force sensor for four motion cycles, corresponding applied voltage, and current of the motor for human upper arm.

probably caused by the tagging pulse of the MR scanner, the force variation can be observed clearly. Although more extensive studies would be needed, the results verified the applicability of the proposed methodology and the system in biomechanical measurements.

V. Conclusion

This paper described a methodology and the corresponding system to visualize dynamic deformation of soft samples. The system employed an electrostatic linear motor that has a capability of open-loop positioning control. For monitoring pushing force generated by the motor, shielded strain gauges were used.

First, the motor motion was confirmed in a normal environment to make sure that the motor can realize constant cyclic motion under resistance from the pushed sample. The results confirmed that the motor can move without step-out even when it pushes the gel sample. Then, the influence of the system on MR images was evaluated. SNR was degraded to a certain extent but no distortion was observed.

Finally, the developed system was applied to the visualization of deformation of a gel sample and a human upper arm. The resultant chronological images clearly showed the deformation of those samples and the force sensor output showed the corresponding force variation. Although it was not the scope of this paper, we think that these results can be used in biomechanical modeling to identify the distribution of visco-elastic parameter, in the future work.

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